

Scheduler Startup Guide

Welcome to the **Scheduler** project! This guide provides step-by-step instructions to get your local environment set up, configure your services, and run both the Python FastAPI backend and the Next.js React frontend.

Prerequisites

Ensure you have the following installed on your machine:

1. **Node.js** (v18 or higher) and **npm**
 2. **Python** (v3.10 or higher)
 3. **uv** (A fast Python package installer and resolver. Install via `pip install uv` if not present)
 4. **PostgreSQL** (running locally on port `5432` with a database named `timetable`)
-

Configuration & Setup

1. Database Setup

Make sure PostgreSQL is running on your system. You must create a database named `timetable` if it does not already exist.

For example, via `psql`:

```
CREATE DATABASE timetable;
```

2. Environment Variables

Backend Configuration (`backend/.env`)

Create or edit the `backend/.env` file in the `backend/` directory. You can copy the template from `backend/.env.example`:

```
# Example backend/.env
APP_ENV=development
APP_PORT=8000
DATABASE_URL=postgresql://<username>:<password>@localhost:5432/timetable
JWT_SECRET=local-development-secret-change-before-production
JWT_EXPIRES_IN_HOURS=168
CORS_ORIGINS=http://localhost:3000,http://localhost:3001,http://localhost:5173
```

Note: Make sure to replace `<username>` and `<password>` with your PostgreSQL credentials.

Frontend Configuration (frontend-next/.env.local)

Create or edit the frontend-next/.env.local file inside the frontend-next/ directory:

```
# Example frontend-next/.env.local
NEXT_PUBLIC_API_BASE_URL=http://localhost:8000/api/v1
```

 Running the Project

Convenient root-level commands are provided in the main package.json to make running the project easier.

Step 1: Install Dependencies

Run the following commands in the root directory:

```
# Install backend dependencies (uses uv sync)
npm run backend:install

# Install frontend dependencies (npm install in frontend-next)
npm run frontend:install
```

Step 2: Run Database Migrations

Run migrations to set up the tables in your PostgreSQL database:

```
npm run backend:migrate
```

Step 3: Start Development Servers

Start both backend and frontend development servers.

```
# In Terminal 1: Start the Backend API
npm run backend:dev

# In Terminal 2: Start the Frontend App
npm run frontend:dev
```

 Useful URLs & Endpoints

Once the servers are running, you can access the following services:

Service	URL	Description
---------	-----	-------------

Service	URL	Description
Frontend Web App	http://localhost:3001	Next.js visual interface
Backend API	http://localhost:8000/api/v1	Main REST API base
Interactive API Docs	http://localhost:8000/docs	Swagger UI for exploring/testing endpoints
Health Check	http://localhost:8000/api/v1/health	Endpoint showing backend & DB health status

Troubleshooting & Verification

- **Verifying Backend Health:** You can run `curl http://localhost:8000/api/v1/health` or open it in your browser. It should return a success JSON indicating the service status is "ok".
- **Database Connection Failure:** Check that your `DATABASE_URL` matches your local database credentials and that the database `timetable` exists.
- **Port Conflict:** If port `8000` or `3001` is already in use, you can modify the ports in `backend/.env` or run Next.js on a different port using the `-p` parameter in `frontend-next/package.json`.